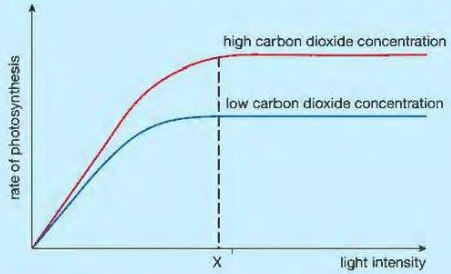
UNIT QUESTIONS

SKILLS ANALYSIS (6)

1 Light intensity and the concentration of carbon dioxide in the atmosphere influence the rate of photosynthesis.

a The graph shows the effect of changing light intensity on the rate of photosynthesis at two different carbon dioxide concentrations.



i Describe the effect of light intensity on the rate of photosynthesis at each concentration of carbon dioxide up to light intensity X and beyond light intensity X. **(4)**

ii Which factor limits the rate of photosynthesis up to light intensity X and beyond light intensity X? **(2)**

Explain your answer in each case. **(3)**

SKILLS CRITICAL THINKING (5) (6)

b iii Describe two other factors which influence the rate of photosynthesis. **(2)**

ii Explain why each is a limiting factor. **(4)**

SKILLS INTERPRETATION

c 'Photosynthesis is a means of converting light energy into chemical energy.' Explain what this statement means. **(2)**

Total 17 marks

SKILLS DECISION MAKING

2 In an investigation to determine the water potential of potato cells, the following procedure was adopted.

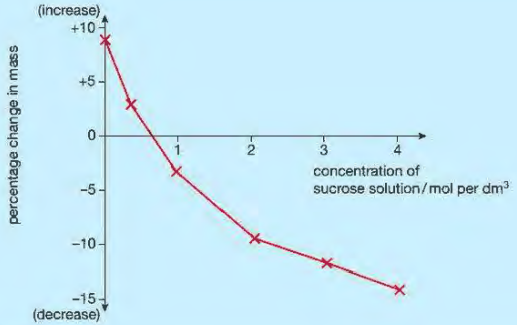
Cylinders of tissue were obtained from a potato. Each was the same diameter and cut to a length of 5 cm.

Each cylinder was dried and then weighed.

Three potato cylinders were placed in each of six different concentrations of sucrose solution and left for two hours.

The cylinders were then removed from the solutions, dried and reweighed. The percentage change in mass for each was calculated, and then an average percentage change in mass calculated for each solution.

The graph summarises the results.



SKILLS DECISION MAKING (5) (4) (6)

a Explain why:

i the cylinders were dried before and after being placed in the sucrose solutions **(1)**

ii three cylinders were used for each solution **(1)**

iii all the cylinders were the same diameter and were cut to the same length. **(1)**

SKILLS PROBLEM SOLVING (9) (8)

b i In terms of water potential, explain the result obtained with a 3 mol per dm³ sucrose solution. **(3)**

ii What concentration of sucrose has a water potential equivalent to that of the potato cells? Explain your answer. **(3)**

SKILLS EXECUTIVE FUNCTION (9)

c How could the water potential of the potato tissue be determined more accurately? **(2)**

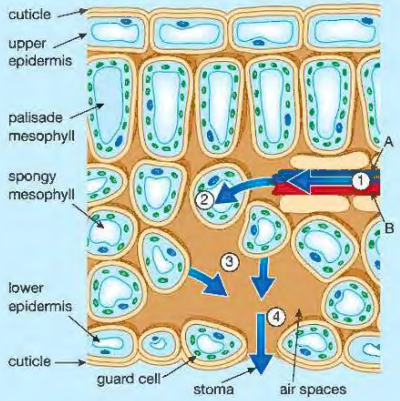
Total 11 marks

BIOLOGY ONLY

SKILLS ANALYSIS

3 Transpiration is the process by which water moves through plants from roots to leaves. Eventually, it is lost through the stomata.

The diagram shows the main stages in the movement of water through a leaf.



SKILLS ANALYSIS (6)

a Name the tissues labelled A and B. Explain how you arrived at your answers. **(4)**

(8) **b** Describe how water is being moved at each of the stages 1, 2, 3 and 4. **(4)**

(5) **c** Describe two ways, visible in the diagram, in which this leaf is adapted to photosynthesise efficiently. **(4)**

(6) **d** For plants living in dry areas, explain a possible conflict between the need to obtain carbon dioxide for photosynthesis and the need to conserve water. **(2)**

Total 14 marks

END OF BIOLOGY ONLY

SKILLS CRITICAL THINKING (5) **4** Plants can respond to a range of stimuli.

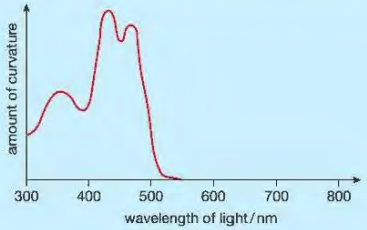
a Plant shoots detect and grow towards light.

i What is this process called? **(1)**

(8) **ii** Explain how a plant bends towards the light. **(3)**

(6) **iii** Explain the advantage to the plant of this response. **(1)**

b In an investigation, young plant shoots were exposed to light from one side. The wavelength of the light was varied. The graph summarises the results of the investigation.



SKILLS ANALYSIS (6) **i** Describe the results shown in the graph. **(2)**

(7) **ii** Suggest why the results show this pattern. **(2)**

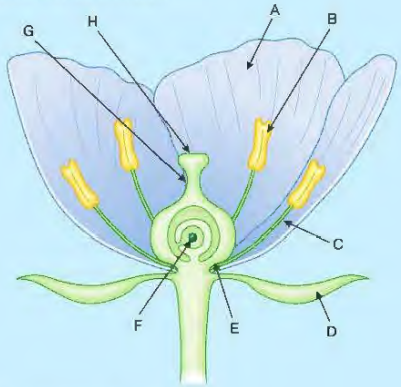
SKILLS CRITICAL THINKING (5) **c** **i** Name another stimulus that produces a growth response in plants. **(1)**

ii Describe the ways roots and shoots respond to this stimulus **(2)**

(6) **iii** What is the benefit to the plant of these responses? **(2)**

Total 14 marks

SKILLS ANALYSIS (4) **5** The diagram below shows a flower.



a Give the letter of the structure which:

i produces pollen grains

ii becomes the seed

iii becomes the fruit wall. **(3)**

b Write down two ways that you can see in the diagram that this flower is adapted for insect pollination. **(2)**

c Give the letter of the structure which is:

i the stigma

ii the style

iii the filament. **(3)**

SKILLS INTERPRETATION (5) **d** Explain the difference between:

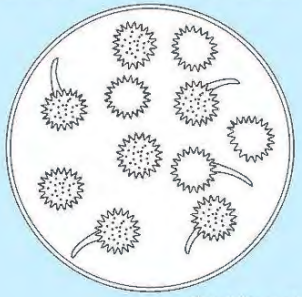
i pollination and fertilisation

ii self-pollination and cross-pollination. **(4)**

Total 12 marks

SKILLS EXECUTIVE FUNCTION (8) **6** When pollen grains land on the stigma of a plant they germinate (grow a pollen tube). Growth of the pollen tube is thought to be stimulated by sugars produced by the stigma.

If pollen grains are placed in certain solutions they will germinate. The drawing shows some grains seen through a microscope after 2 hours in a solution.



Design an investigation to find out if germination of pollen grains is stimulated by sucrose.

Your answer should include experimental details and be written in full sentences.

Total 6 marks